

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Description	Module	Status
1	Model Training Pipeline	Trains & compares 4 ML models, saves the best one	train.py	Done
2	Flood Prediction API	Returns flood probability for submitted weather data	app/routes.py, app/ml.py	Done
3	Dynamic Result Display	AJAX-rendered risk badge and probability bars, no reload	static/js/script.js	Done
4	Prediction History Logging	Every prediction saved to the database automatically	app/routes.py, app/models.py	Done
5	History Page	Table view of the most recent predictions	templates/history.html	Done
6	Auto-Generated Insights	EDA and model-comparison charts shown on the home page	train.py, templates/home.html	Done
7	Responsive UI	Mobile and desktop layouts with a blue/green theme	static/css/style.css	Done
8	Cloud Deployment	Live, publicly accessible deployment via Render	Render Web Service	Done

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (training, API, UI form, dynamic results, DB logging, deployment)
Additional / Nice-to-Have Features	2 (history page, model insight charts)
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page, prediction form, history table
2	Backend / Logic	2	Flask routes, ML inference module
3	Database / Storage	1	SQLite models (User, WeatherData, MLModel, PredictionResult)
4	API / Integration	1	POST /api/predict JSON endpoint
5	Security / Authentication	1 (partial)	Guest-user logging; schema supports full auth later